

```

10 iterations: 3.1424259850010982653: Percent Error:
0.026525762668591974791
100 iterations: 3.1416009869231253937: Percent Error:
0.00026525823845289050367
1000 iterations: 3.1415927369231226862: Percent Error:
2.6525822650814072353e-06
10000 iterations: 3.1415926544231336237: Percent Error:
2.6526052216183079483e-08
100000 iterations: 3.1415926535981619772: Percent Error:
2.6638912432079985618e-10
999999999 iterations: 3.1415926535898006655: Percent Error:
2.403085759327990643e-13

```

What factors affect the accuracy of the value of π you calculate?

- The main factor affecting the calculation is the number of iterations N of the Riemann Sum. As N increases, the absolute error decreases and as N approaches infinity, this sum becomes exact.
- There is also computational expense associated with larger and larger N values

What is the most accurate value of π you can get, and what is the corresponding value of N ?

- 3.1415926535898006655 is the value of π I calculated.
- This is with a percent error of $2.4e-13\%$, which was after 999,999,999 iterations. This error is so small that it is approaching machine precision for doubles.
- In theory, you can run this program for a larger value of N and reach machine precision, but this is computationally very expensive.